NBA Data Warehouse

Project Summary and Data Warehouse Documentation

PostgreSQL 16 | Docker Compose | SQL | Star Schema

1. Project Overview

The NBA Data Warehouse project organizes historical NBA game records into a structured PostgreSQL database. It uses an Extract, Transform, and Load (ETL) process to prepare source data for querying and analysis.

The warehouse follows a Star Schema design, with a central game fact table connected to date, team, arena, and game type dimensions.

2. Dataset Summary

Metric	Value
Total games	**73,470**
Teams	**71**
Arenas	**276**
Game types	**8**
Date dimension rows	**29,200**

Dataset Detail	Value
First game date	November 26, 1946
Last game date	November 5, 2026

Average home score	106.06
Average away score	102.52
Average attendance (available values only)	17,851

3. Database Design

Fact Table

`dw.fact_game` stores game-level records, including game ID, date and dimension keys, scores, winner, and attendance.

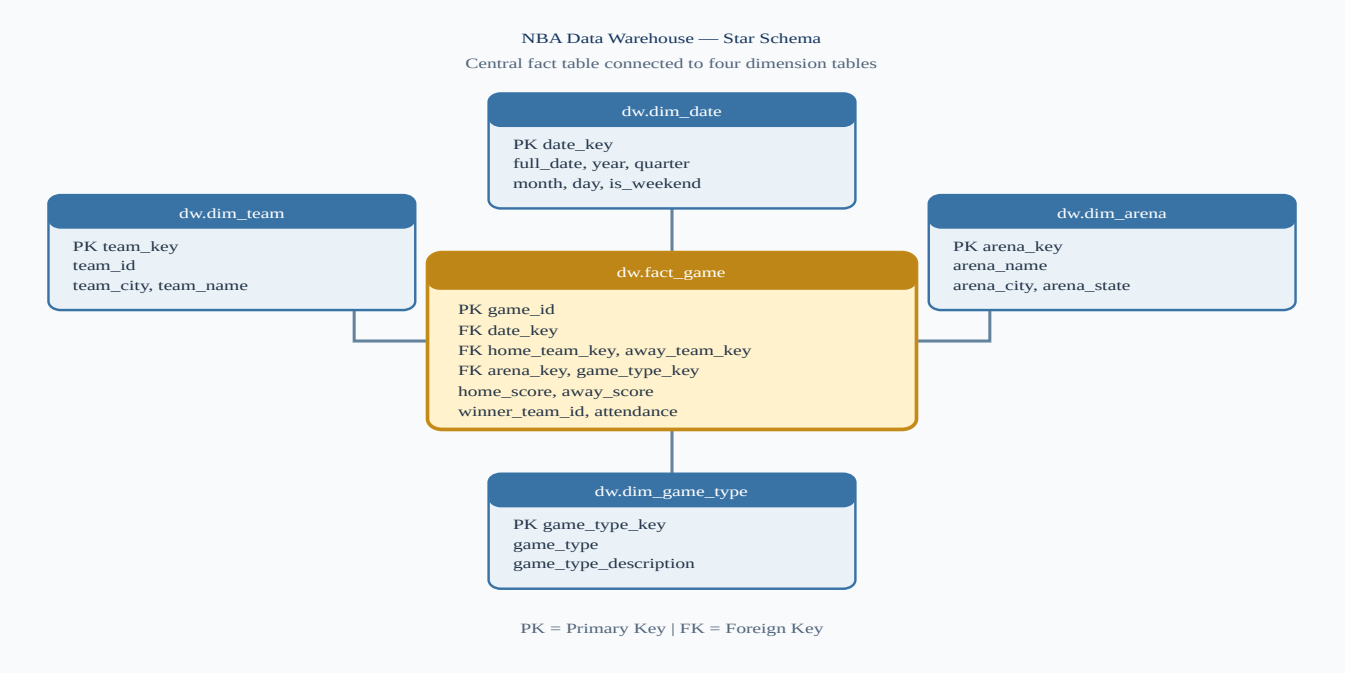
Dimension Tables

Dimension Table	Description
`dw.dim_date`	Date attributes
`dw.dim_team`	Team information
`dw.dim_arena`	Arena information
`dw.dim_game_type`	Game type information

The staging schema (`stg`) holds source records, while the ETL schema (`etl`) holds cleaned and transformed data.

4. Star Schema Diagram

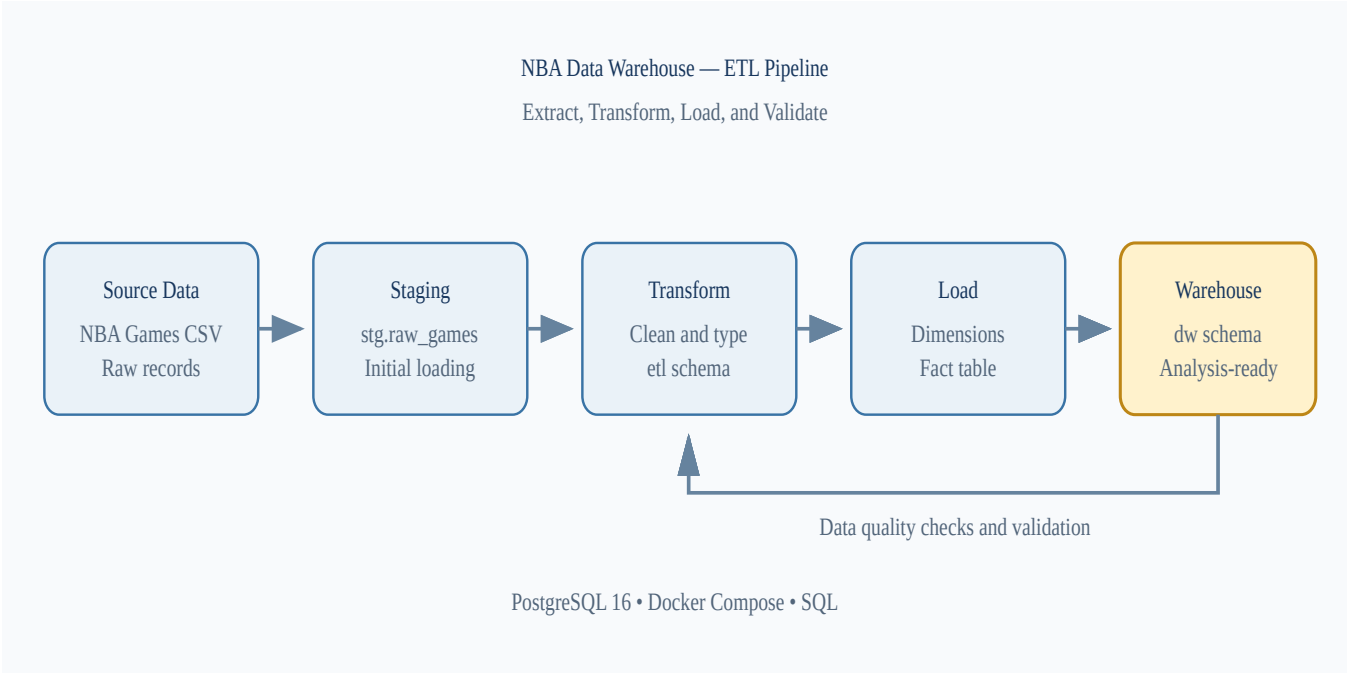
The fact table is connected to four dimension tables: date, team, arena, and game type.



5. ETL Pipeline

The ETL process follows these stages:

Stage	Process	Database Location
1. Source	NBA game records in CSV format	Source file
2. Extract	Load raw records	`stg.raw_games`
3. Transform	Profile, clean, and convert data types	`etl` schema
4. Load	Load dimension and fact records	`dw` schema
5. Validate	Check row counts and data quality	Validation queries



6. ETL Row Counts

ETL Stage	Number of Rows
`stg.raw_games`	73,470
`etl.cleaned_games`	73,470
`etl.typed_games`	73,470
`etl.typed_games_final`	73,470
`dw.fact_game`	73,470

> All checked ETL stages contain the same number of rows.

7. Data Quality Summary

Data Quality Check	Result
Total fact rows	73,470

Unique game IDs	73,470
Duplicate game IDs	0
Missing checked critical fields	0
Invalid negative scores	0
Missing attendance values	71,887

> Attendance is incomplete in the source data. Average attendance is calculated only from records with available attendance values. Missing attendance is not treated as zero.

8. Conclusion

The project transforms historical NBA game records into a structured PostgreSQL data warehouse. The Star Schema supports organized data queries, while the ETL process prepares records for analysis. Validation checks found no duplicate game IDs or negative scores among the checks reported in this document.

9. Project Documentation

Document	File
Lab 1 Documentation	LAB_1_DOCUMENTATION.md
Lab 2 Documentation	LAB_2_DOCUMENTATION.md
Project README	README.md